Effect of cell growth supporting surface on phagocytosis

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Abstract

Phagocytosis is involved in several functions including nutrient uptake, immune response, inflammation, tissue homeostasis, cellular apoptotic bodies and debris elimination. Its dysfunction has impacts on normal functioning of an organism. It is involved in several diseases including the AMD pathogenesis, and treatment approaches. Mostly an initial proof of concept is tested in a laboratory conditions. Cell and tissue cultures need proper surfaces to grow and maintain them in laboratory conditions. The study aims to evaluate effect of two surfaces; tissue culture plate polystyrene and glass coverslip, on phagocytosis performed by RPE cells growing on each surface. Fresh RPE cells and ARPE-19 cell line were grown in corresponding cell culture conditions. Phagocytosis in fresh RPE cell cultures was detected and recorded. ARPE-19 cells were incubated for 4 and 24 hours with 0.2 and 1 µl red fluorescent latex beads, and then images were taken to measure the average numbers of cells, cells performing no phagocytosis and phagocytosis and engulfed red fluorescent latex beads. Results showed that RPE cell membrane was forming blebs releasing black dots like particles as cell debris in cell culture medium. Subsequently cytoplasmic membrane extensions were formed which had engulfed black dots like particles. ARPE-19 cells are a good cellular model for performing initial in vitro test in comparison to fresh RPE cells. There was no significant difference between numbers of ARPE-19 cells growing on polystyrene and glass surface performing latex bead phagocytosis. Both had engulfed red fluorescent latex beads; however, numbers of engulfed beads was significantly higher for glass surface than polystyrene surface. Results support the hypothesis that surface of cell culture plates affects the cellular phagocytosis mechanism, which play a critical role in the functions in many cases such as in vivo retina. These findings are very crucial when it comes to transfer a treatment approach based on phagocytosis to the clinics and that the materials and the conditions of the experiments can have very important consequences

Keywords

Retinal pigment epithelial cells, phagocytosis, cell growth supporting surface, latex beads

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Introduction

Phagocytosis is a cellular mechanism by which cells engulf large particles (> 0.5 micrometers), microorganisms, pathogens, dead or dying cells, cell debris from their extracellular space. The engulfed particles are internalized within a plasma membrane derived vesicle named phagosome, and then they fuse with lysosomes for enzymatic degradation. The mechanism of phagocytosis can be divided into four main steps: (i) target particles recognition, (ii) internalization machinery signaling pathway activation, (iii) phagosome formation, and (iv) phagolysosome maturation (Rosales and Uribe-Querol, 2017). The monocytes, macrophages, neutrophils, dendrictic cells, osteoclasts, eosinophils are some of the cells able to phagocitize (phagocytes). They are involved in eliminating microorganisms and presenting them to cells of the adaptive immune system (Rabinovitch, 1995). Besides, fibroblasts, epithelial cells and endothelial cells also perform phagocytosis including eliminating apoptotic bodies and cellular debris. In unicellular organisms, the phagocytosis mechanism supports to nutrient uptake, however in multicellular organisms they are involved in tissue development remodeling and homeostasis, immune response and inflammation (Rosales and Uribe-Querol, 2017). Failure of phagocytosis has impact on normal functions of an organism including the pathogenesis of age related macular degeneration (AMD) (Li, 2013; Rosales and Uribe-Querol, 2017).

Retinal pigment epithelium (RPE) is a monolayer of highly specialized cells situated between neural retina and Bruch's membrane. They interact at their apical surface with photoreceptors (PR). On its apical and basal surface this monolayer surface performs many functions including light absorption, epithelial transport, glia, visual cycle, photoreceptor waste recycling, and bioactive molecules secretion (Sparrrow et al., 2010). The apical surface performs phagocytosis of photoreceptor outer segments discarded for their renewal. In in vivo condition, throughout whole life, the photoreceptors undergo daily renewal and shedding of outer segment (OS) discs. The RPE engulfs these discarded discs, performs their enzymatic degradations in cellular lysosome system, and then release metabolism residues on its basal surface and further these residues are transported through choriocapillary. If somehow this mechanism is blocked, OS discs degraded debris accumulates on basal surface which affects normal functions of RPE (Li, 2013). This produces an influence on the health of different retina layers including photoreceptors, and which could lead into generating retinal diseases such as retinal degeneration, retinitis pigmentosa, and above all age related macular degeneration which are of significant clinical interest (Kocaoglu et al., 2016; Nguyen-Legros and Hicks, 2000). Hence it is crucial that RPE maintains its phagocytosis mechanism in normal function.

Adult and stem cells are currently some of the best options to treat many diseases. They are either injected directly into damaged tissue of the patient or propagated in lab conditions using cell culture techniques before injection (Nadig, 2009). Some of them undergo for regenerative medicine procedures before placing into damaged tissue of the patient (Dąbrowska and Skopiński, 2017). For example, they are grown on a suitable biomaterial surface for making a cell-biomaterial scaffold which is placed to repair the damaged tissue. Nevertheless cells cannot avoid the roles of genetic factors and ageing process (Rodríguez-Rodero et al., 2011). The adult stem cells based therapies use two characteristics; neuroprotection or a target cell differentiation. The differentiated cells are used to repair the damage tissues of the patients. Several stem cells derived differentiated RPE cells are in clinical trials phase I and II (Trounson and McDonald, 2015). Nevertheless pushing adult cells or stem cells based therapy towards a clinical application need several tests for establishing the proof of concept (PoC). It begins with easy acquisitions of healthy human tissue, cell isolation, cell culture, and cell propagation, application of procedures of targeted treatment strategy, phenotyping and functional characterizations for measuring targeted treatment strategy. Likewise phenotype measurement,

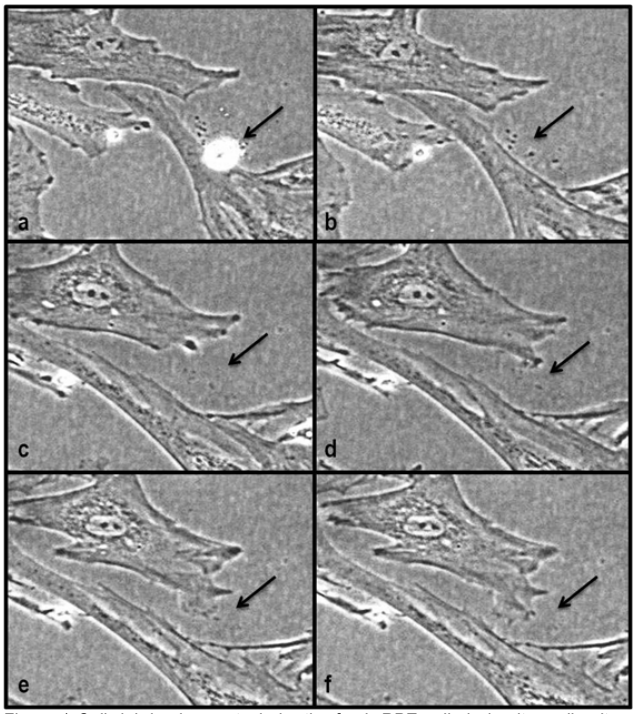


Figure 1 Cell debris phagocytosis by the fresh RPE cells in in vitro cell culture conditions. Image a) represents cell blebbing, Image b) represents the cell debris (black dot particles) left after a cell blebbing, Images, c) to f), represents cell membrane prolongations to engulf cell debris. Arrows are pointing cell blebbing, cell debris and cell membrane and cytoplasm extensions engulfing cell debris.

functional characterizations are unavoidable for correct functioning of a treatment (Haruta et al., 2004; Phelan and May, 2016). Cells are grown on different surfaces such as synthetic or natural polymers or a mixture of both polymers for cell culture propagation and then applying cell and regenerative medicine based strategies. There are a variety of surfaces which have been studied and reported for RPE cells. Some of them are a substrata developed from microspheres with cross-linked fibrinogen, amniotic membrane, anterior lens capsule, Descemet's membrane, cryoprecipitated extracellular matrix membranes, cadaver Bruch's membrane, polystyrene for tissue culture coated with either collagen, Matrigel™, poly(d-lysine) and ECM components, poly(α-hydroxyesters), polylactides, polyglycolides and their copolymers, poly(l-lactic acid) (PLLA) and poly(lactic-co-glycolic acid) (PLGA), poly(dl-lactic acid)/poly(ethylene glycol) (PLA/PEG), poly(dl-lactide-co-glycolide), poly(ε-caprolactone) (PCL), Poly(3-hydroxybutyrate-co-3-hydroxyvalerate), laminin-coated poly(methyl methacrylate) (PMMA), hydrogel based on methacrylamide, coated with poly(d-lysine) and fibronectin, polyurethanes (Lu et al., 2001).

In laboratory conditions, two of popular surfaces are polystyrene surface of tissue culture plates treated for cell and tissue cultures, and glass surface of cover slips for performing cell and tissue culture, and then performing further experimental studies. The purpose of this work has been to analyze the possible influence of these two surfaces in the phagocytosis capacity of surface bound-growing RPE cells.

Materials and methods

Retinal pigment epithelial cells

The RPE cell cultures were prepared for phagocytosis study as has been described in our previous papers (Rodriguez-Crespo et al., 2014; Srivastava et al., 2013). In brief, a cell line of human RPE cells (ARPE-19) was purchased from the American Type Culture Collection (ATCC) and grown in DMEM/F12 cell culture medium. The fresh RPE cells were isolated from human eye globes and grown in DMEM cell culture medium. The standard cell culture conditions of both cultures were incubation of cells in corresponding cell culture medium in an incubator regulated for 5% CO2 and 37ºC. The cell culture medium was supplemented with a 1% antibiotics- antimycotic solution (Gibco®, Paisley, UK) for inhibiting bacterial and fungal contaminations and 10% fetal bovine serum (FBS) (Gibco®, Paisley, UK) for providing growth factors for cell growth. The cell culture medium was renewed at every 2-3 days with the fresh cell culture medium for sustain nutrients supply for cell growth. The cell culture was serially propagated by sub-culturing technique using 0.05% Trypsin-EDTA solution (Gibco®, Paisley, UK) for trypsinization and 25 to 75 cm flasks for sub-culturing. The single cell suspensions of the cells were prepared after trypsinization of >90% confluent cell cultures. The trypan blue exclusion assay was performed to measure the cell viability in cell suspensions. The cell suspensions showing >90% cell viability were seeded in a cell density of 1 x 10^4 cells per well for performing subsequent experimental works.

***In vitro* phagocytosis:**

Fresh RPE cells were seeded in a 35 mm Petri dish (Thermo Scientific™) and grown in standard cell culture conditions. The semi confluent cell cultures were observed under an inverted microscope (Axiovert 100 Series microscope of Carl Zeiss). The cellular mechanisms undergoing in in vitro cell cultures were recorded during 12 to 24 hours.

Red fluorescent latex beads phagocytosis

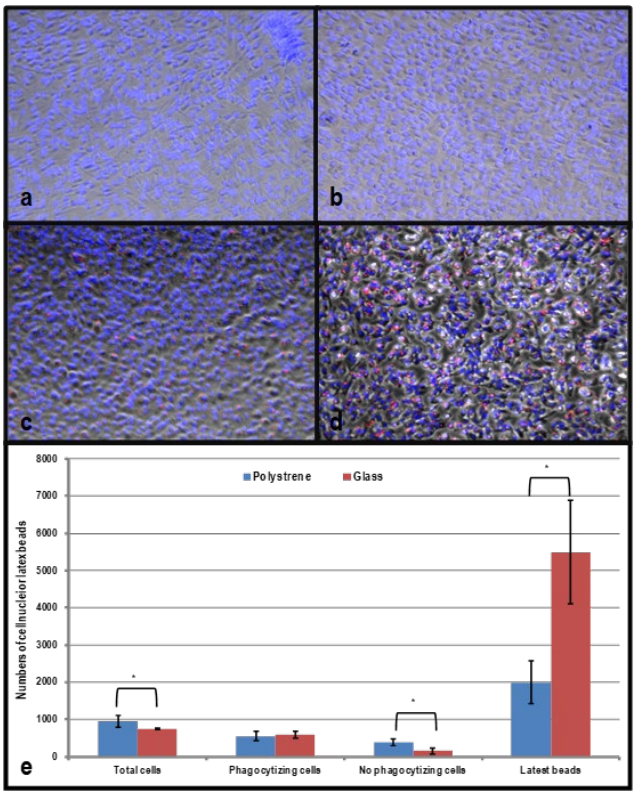


Figure 2 Red fluorescent latex beads phagocytosis (1 µl) by ARPE-19 cells growing on different surfaces; a) and c) cell culture polystyrene surface b) and d) glass coverslip surface. Images a) and b) represent control cell cultures without incubations with 1 µl of red fluorescent latex beads. Images c) and d) represent cell cultures incubated for 4 hours with 1 µl of red fluorescent latex beads for phagocytosis. Cells of each group were observed under an objective lens of 10X magnification of a microscope LEICA DMI6000 B. Cells growing on cell culture polystyrene surface had engulfed less red fluorescent latex beads than those were growing on glass coverslip surface. Image e) represents average numbers of nuclei and engulfed latex beads and related standard deviations on each surface on counting 5 images per surface using ImageJ software.

Phagocytosis study was performed in 24-well flat-bottomed cell culture plates (Costar, Corning Inc., Corning, NY, USA). These cell culture plates are made of polystyrene, hence each well had a surface of polystyrene for cell growth. The circular glass cover slip (diameter: 12 mm; Thermo Scientific) was inserted in 24-well cell flat-bottomed culture plates to provide a surface of glass to cell growth. Then plates were exposed to UV light for overnight and then they received

DMEM cell culture medium supplemented with antibiotics for 2 hours for sterilization before cell seeding. The single cell suspensions of the ARPE-19 cells were prepared after trypsinization of >90% confluent cell cultures. The trypan blue exclusion assay was performed to measure the cell viability in cell suspensions. The 1 x 10^4 cells showing ~ 90% viability were seeded and grown in each well of 24-well flat-bottomed cell culture plate on the surfaces of polystyrene and glass in standard cell culture conditions of 5% CO2 and 37ºC. When cell cultures achieved ~ 80% to 90% confluence level as confirmed by a microscope (Phase contrast - fluorescence microscope DMI6000 B, Leica Microsystems, Mannheim, Germany), the entire medium was removed from each well and cell cultures were washed with cell culture medium to remove non-adhered cells. Following they were incubated for 4 and 24 hours with 0.2 and 1 µl red fluorescent latex beads (Sigma Aldrich, L2778) in cell culture medium. Subsequently the cell cultures were washed to remove extracellular latex beads. Cell cultures were also stained with DAPI (Invitrogen, Eugene, OR) to make visible the cell nuclei (Rodriguez-Crespo et al., 2014). Immediately cell cultures were observed under objective lens of 10X, 20X and 40X magnifications of a microscope LEICA DMI6000 B and images were photographed for each surface at random. The Images were analysed using ImageJ software for counting the red fluorescent latex beads and DAPI stained blue nuclei. A bar graph was created for average numbers of cell nuclei and engulfed latex beads on each surface, with related standard deviations on counting 5 images per surface using ImageJ software. Similarly the manual countings of red fluorescent latex beads present in the cytoplasm of a selected cell was done to measure maximum numbers of red fluorescent latex beads engulfed by a cell growing on each surface.

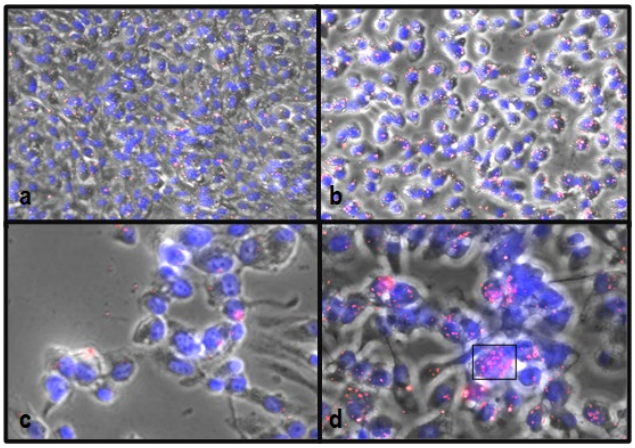


Figure 3 Red fluorescent latex beads phagocytosis by growing ARPE-19 cells on different surfaces; a) and c) cell culture polystyrene surface b) and d) glass coverslip surface. All groups were incubated for 4 hours with 1 µl of red fluorescent latex beads for phagocytosis. a) and b) represent the cells observed under an objective lens of 20X magnification, c) and d) under 40X magnification of a microscope LEICA DMI6000 B. Cells growing on cell culture polystyrene surface had engulfed less numbers of red fluorescent latex beads than those were growing on glass coverslip surface. In rectangular area of image d) it was counted manually 30 latex beads which a cell had engulfed.

Results

Cell debris phagocytosis

The cell body, cell nucleus and its membrane, cytoplasm, pigments, cell blebbing, cell debris, cell membrane and its extension, phagocytosis of cell debris in in vitro fresh RPE cell cultures were observed in microscopic recorded images (figure 1). The image 1 a) and b) showed cell debris (black dots like particles) which were left in the cell culture medium on cell blebbing (arrow pointing a cell bleb and debris left). The images, 1 c) to f), showed the cell membrane and cytoplasm extensions which were engulfing the cell debris. The videos 1 clearly demonstrated a fresh RPE cell executed phagocytosis for cell debris engulfing and clearance, and video 2 demonstrated the cell division in in vitro cell culture conditions. The results confirmed that fresh RPE cells did not lose their phagocytizing capacity in in vitro conditions, however, it would be important to study if there is any change in this capacity due to a growth supporting surface using a suitable cellular model (APRE-19).

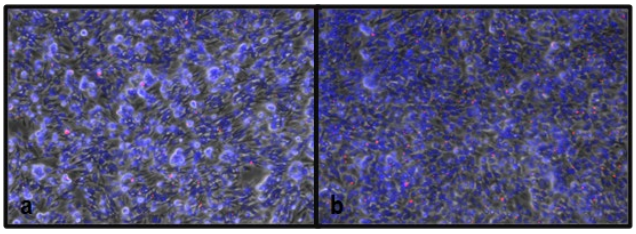


Figure 4 The 0.2 µl of red fluorescent latex beads phagocytosis by growing ARPE-19 cells on different surfaces; a) cell culture polystyrene surface b) glass coverslip surface. Images a) and b) represent cells after incubation for 4 hours with 0.2 µl of red fluorescent latex beads. The cells of each group were observed under an objective lens of 10X magnification of a microscope LEICA DMI6000 B. Cells growing on cell culture polystyrene surface had engulfed less red fluorescent latex beads than those were growing on glass coverslip surface concluding consistency of results observed in figure 2 and 3.

Latex bead phagocytosis

ARPE-19 cells grew well on cell culture polystyrene surface and glass surface in standard culture conditions (figure 2). Microscopic observations did not show any significant alterations in cell morphology on both surfaces. Staining with the DAPI demonstrated the presence of round to oval shaped nuclei on both surfaces. Cells were incubated for 4 hours with 1 µl of red fluorescent latex beads for phagocytosis in cell culture medium. The cells growing on the polystyrene and glass surface showed no red fluorescent particles in cell cytoplasm when they were not incubated with red fluorescent latex beads (figures 2 a and b). However, when they were incubated with latex beads, they showed red fluorescent particles i.e. latex beads in cell cytoplasm (figures 2 c and d). The microscopic images showed that cells growing on glass surface had engulfed more latex beads than those growing on the polystyrene surface (figures 2 c and d). On counting and further data analysis showed that there were 888±115 nuclei on polystyrene surface and 966±22 on glass surface (figure 2 e). The numbers of red fluorescent latex beads were 1084±301 on polystyrene surface and 1242±126 on glass surface (figure 2 e). There were differences on comparison among average numbers for different groups; between total numbers of nuclei and engulfed red fluorescent latex beads on polystyrene surface ($p=0.211$) and glass surface ($p=0.005$), between polystyrene and glass surface for total numbers of nuclei ($p=0.231$) and engulfed red fluorescent latex beads ($p=0.366$), between total numbers of nuclei on glass surface and engulfed red fluorescent latex beads on polystyrene surface ($p=0.465$) and between total numbers of nuclei on polystyrene surface and engulfed red fluorescent latex beads on glass surface ($p=0.003$). The p value calculated for differences between different groups showed that differences were significant only in two cases ($p \leq 0.05$) (figure 2 e). The comparisons of images of figures 3 a of 20X and c of 40X microscopic magnification for cells growing polystyrene surface with the similar images, figures 3 b and d, for cells growing glass surface showed previous observations and results. The images of figures 3 c and d also showed that all of the engulfed red fluorescent latex beads were inside the cell cytoplasm. No one was present in cell culture medium. Maximum 30 red fluorescent latex beads were engulfed by the cell growing on glass surface (figure 3 d). However, for polystytene surface it was 5 (figure 3 c). To confirm further that the level of cell phagocytosis capacity was altered due to surface on which cells were growing, the cells were incubated with 0.2 µl of red fluorescent latex beads for 4 hours. The results again showed the same observations and results which were found in previous experiments. The cells growing on polystyrene surface engulfed less numbers of red fluorescent latex beads than those growing on glass surface (figure 4 a and b). To observe the time dependent cell capacity to engulf the red fluorescent latex beads, the cells growing on polystyrene surface were incubated for 24 hours with the 1 µl of red fluorescent latex beads. The results showed that the numbers of cells engulfing the red fluorescent latex beads was much higher on 24 hours than 4 hours. However, it was not observed a big change in maximum numbers of red fluorescent latex beads engulfed by a cell on this surface on comparing the images obtained on both time points (figure 3 c and figure 5 a).

Discussion

Fresh healthy adult cells from patients propagated in lab conditions, and stem cell differentiated cells are very useful in applying cell therapy and regenerative medicine therapy to treat the patients. The phenotype and functional characterisations of these cells are crucial in deciding a treatment will work in the patients. In regenerative medicine studies, a number of polymers are proposed, tested or under tests as cell matrices, and some of them are already in regular clinical applications (Bhat and Kumar, 2013). Nevertheless, mostly in laboratory conditions, cell culture plates which provide treated polystyrene surface, and glass coverslip are used for cell and tissue cultures. The phenotypes of a cell are normally well studied during investigation but it could be crucial to know the impact of surface on functional characteristics of the cell. This makes to perform a study for measuring the functional characteristics to select a proper surface for cell culture propagation.

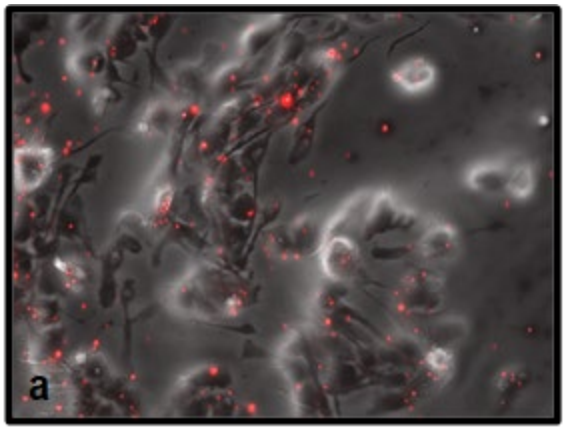


Figure 5 The 1 µl of red fluorescent latex beads phagocytosis by ARPE-19 cells growing on cell culture polystyrene surface. a) a representative image at 40X microscopic magnification of the cells after incubation for 24 hours with 1 µl of red fluorescent latex beads.

One of the critical roles of RPE cells in vision is performed by phagocyting the outer segments of the photoreceptor which are discarded for their renewal during whole life time (Kevany and Palczewski, 2010). Phagocytic function of a human RPE cell line, ARPE-19, possesses similar phagocytic machinery as RPE in situ (Mao and Finnemann, 2013; Tian et al., 2017). The fresh primary cell cultures of human originated RPE cells show many hurdles to obtain, grow and maintain in a laboratory conditions where an established cell line, ARPE-19 cells, show advantages (Rodriguez-Crespo et al., 2014; Srivastava et al., 2013). Nevertheless, a treatment strategy must be tested with several fresh primary human RPE cell cultures before transferring to perform *in vivo* test. In standard cell culture conditions cells of fresh cell and cell lines cultures properly divided, undergone cellular blebbing, leaving cell debris in cell culture medium, and performing phagocytosis of cell debris (Strunnikova et al., 2004). It was visible by microscope how cell membrane had created cytoplasm extensions towards cell debris to engulf them. It has been seen that the many researchers showed doubts for contaminations in RPE cell cultures due to presence of black dots like particles in cell cultures observed under microscope. This study (images and videos) provide a proof that these black dots like particles are formed during cellular blebbing process in cell cultures and they are not a product of contamination (figure 1, videos 1 and 2). Nevertheless, this will be an issue of study to investigate what these particles contain; pigments, cell membrane actin filaments, portion of cytoplasm are some of them to be considered.

ARPE-19 cells were grown on polystyrene and glass surface and results showed that they grew well on both surfaces in standard cell culture conditions (figure 2 a and b). There were not found any detectable and significant changes in cell morphology. Although it cannot be discarded molecular changes which might occurred in situ of the cells due to influence of each surface. This study was focused on to explore the effect of cell growth supporting surface on phagocytosis. When these growing cells were incubated for 4 hours with 1 µl of red fluorescent latex beads, cells had engulfed them on both surfaces confirming that both surfaces did not block the cellular phagocytosis mechanism. At this point the interesting finding was that the numbers of cells growing on glass surface showed more phagocytosis in comparison to polystyrene surface. Another finding was that the cells growing on glass surface had engulfed significantly more red fluorescent latex beads than those growing on polystyrene. This finding states that a cell growth supporting surface affect the phagocytosis mechanism. Since this study lacked a real control which determines the level of phagocytosis capacity in an *in vivo* condition for red fluorescent latex beads, it is difficult to state that polystyrene surface had reduced or maintained the *in vivo* level of phagocytosis capacity and glass surface had increased or reduced or maintained this level. It is well known that cell culture plates are made of polystyrene and treated to adapt the cell and tissue cultures. However, in most of the cases, these treatments are undisclosed by manufacturing companies. In further microscopic magnifications, 20X and 40X, it was clearly observed the all the engulfed red fluorescent latex beads were inside the cytoplasm of cells. Cells growing on glass and polystyrene surface had engulfed maximum 30 and 5 red fluorescent latex beads per cell during 4 hours of incubation. This again confirmed that cells growing on glass surface are much active for phagocytosis than the cells growing on polystyrene surface. This difference is not bound to concentrations of red fluorescent latex beads used for incubations because results and observations were same either it was used 1 µl or 0.2 µl of red fluorescent latex beads in incubations. To further to explore if this is a time dependent, the cells growing polystyrene surface were incubated with 1 µl of red fluorescent latex beads for 24 hours. The results clearly showed that on 24 hours incubation the numbers of the red fluorescent latex beads was much higher on 24 hours than 4 hours. Nevertheless, it cannot be discarded a fact that these cells are dividing in cell culture and with time their numbers increase in cell culture. At this point an interesting finding was maximum numbers of red fluorescent latex beads engulfed by a cell was similar which was found on 4 hours incubation. This confirmed there is no change in the level of phagocyosis capacity of a cell along with time. Hence, it can be achieved on a conclusion that cell growth supporting surface affects the phagocytosis mechanism which play a critical role in determining the functions in many cases such as *in vivo* retina. These results seem very relevant when it comes to transfer to the clinic (Tian et al., 2017) and that the material and the conditions of the experiments can have very important consequences.

Statistical Analysis

The results obtained for each substrate were expressed as mean number of cells per field of view (x10) ± standard deviation (SD), as calculated using MS excel 2007. Data were tested for normality and investigated for statistical significance using Student's t-test. Statistical significance was set at $p<0.05$.

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